

TermPlanMT: Graph-Grounded Terminology Planning for French–English Regulatory Medical Document Translation

When Ontology Grounding Helps and When It Does Not

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Abstract

Terminology consistency is a patient-safety requirement in regulatory medical translation, yet neural MT systems are structurally indifferent to cross-segment term stability. We present **TermPlanMT**, a six-system ablation study on Section 4.8 of the KEYTRUDA SmPC—a 126-segment French–English corpus with maximum adverse-event density. Our central finding is that Concept Coverage Rate (CCR), the fraction of source spans grounded in a MedDRA ontology graph, is the binding constraint on terminology-aware MT: at 39% CCR graph-free baselines (S1–S2) outperform all graph-augmented systems on BLEU and chrF, while improving NER quality yields ≈ 7 BLEU gains for Mistral graph systems (S3–S5-Mistral) but not for graph-free S1 or S5-NLLB; S2 is condition-invariant (same inference run in both conditions); S3–S5-Mistral are not (Table 5). We further show that our Hierarchy-Aware Terminology Metric (HTM) scores the certified human reference at about 0.15–0.18 depending on NER manifest, a score unchanged by full MedDRA ancestor alias expansion, confirming a structural mismatch between MedDRA nomenclature and professional translation practice. Finally, ambiguous MedDRA grounding causes the planning stage to abstain from locking critical terms, explaining a paradox in which graph-augmented systems produce *lower* cross-document consistency than graph-free baselines. We complement aggregate scores with a glossary-oracle ablation (S6) and segment-level qualitative analysis (missing-term rates, drift, modal agreement) exported under `error_analysis/` in the project repository.

1 Introduction

EMA Summaries of Product Characteristics (SmPCs) require a single adverse-event concept to receive one identical English rendering across an entire document—a requirement that affects pharmacovigilance signal detection and patient safety [Neves et al., 2022]. Neural MT handles medical prose fluently but is structurally indifferent to cross-segment consistency, producing term drift (same source term, multiple renderings), concept flattening (LLT-level terms collapsed to hyponyms), and coverage gaps (multi-token spans outside NER). MedDRA’s five-level hierarchy (Table 1) provides validated English labels that are an attractive grounding resource—if the grounding pipeline can cover enough of the source text.

Contributions.. (1) A six-system ablation ladder (§4) isolating the effects of NER recall, ontology grounding, and decoding enforcement. (2) HTM and alias-expanded HTM-A (§5), with a diagnostic evaluation showing the metric systematically penalises the human reference. (3) A CCR-bottleneck experiment (§6) showing that NER quality—not graph retrieval strategy—determines whether ontology grounding helps. (4) A mechanistic account (§8) of why graph-augmented systems are *less* consistent than graph-free ones. (5) Structured qualitative error analysis tying drift to ambiguous grounding and abstaining locks, with reproducible CSV/MD artefacts.

2 Related Work

Lexically constrained decoding [Hokamp and Liu, 2017, Post and Vilar, 2018] and terminology-aware training [Dinu et al., 2019] enforce surface-level term appearance but ignore ontological hierarchy and do not address candidate selection when multiple levels are valid. Herold and Ney [2023]

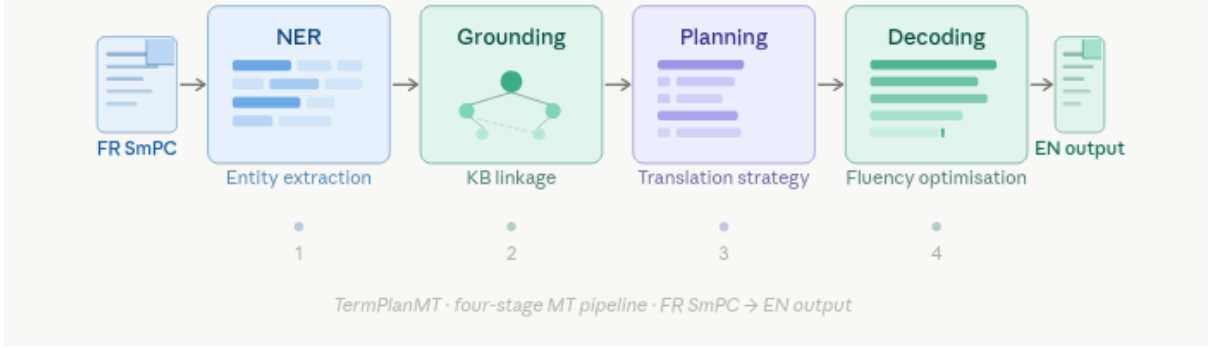


Figure 1: Four-stage TermPlanMT pipeline (NER → grounding → planning → decoding).

HTM — Hierarchy-Aware Terminology Metric

HTM measures how far a translated term has drifted from its source position in the MedDRA hierarchy. Each MT output term is grounded in Neo4j to recover its level $\hat{d}_i$, then compared against the source level d_i :

$$\text{HTM} = \frac{1}{N} \sum_{i=1}^N s_i, \quad s_i = \begin{cases} 1.0 & \hat{d}_i = d_i \text{ (exact level match)} \\ 0.5 & \hat{d}_i \neq d_i, \text{ same branch (wrong specificity)} \\ 0.0 & \text{different branch or ungrounded} \end{cases} \quad (2)$$

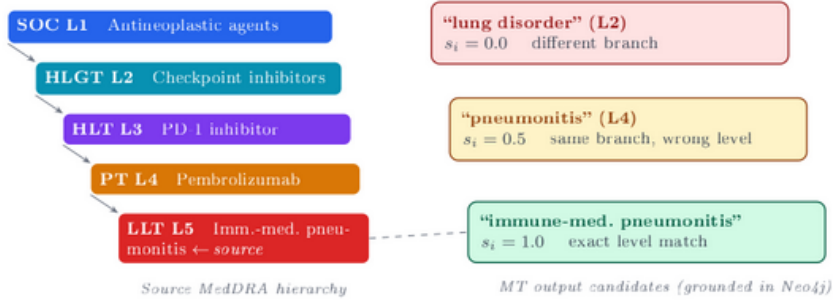


Figure 2: HTM scoring for *pneumopathie immuno-médicamentieuse* (source: LLT, L5). Each MT output is grounded in Neo4j and its MedDRA level compared against the source.

show that extending NMT context windows can degrade fluency, motivating document-level LLMs as a consistency baseline. Labrak et al. [2024] release BioMistral, the medically adapted LLM used for NER here. Neves et al. [2022] identify adverse-event terminology as a high-priority MT failure mode. GraphRAG-style retrieval [Edge et al., 2024] into LLM prompts provides the template for our S3 system, adapting knowledge-grounded MT [Moussallem et al., 2019] to a Neo4j MedDRA backend.

3 Dataset

Corpus.. Section 4.8 (Undesirable Effects) of the KEYTRUDA (pembrolizumab) SmPC [European Medicines Agency, 2024]: 126 aligned French–English segment pairs after excluding segment 48_028 (a dense frequency table whose n-gram overlap distorts corpus metrics). Coverage:

≈4,400 words; all three failure modes at high density.

Level	Abbr.	Example
L1	SOC	Respiratory disorders
L2	HLGT	Pulmonary conditions
L3	HLT	Pneumonitis
L4	PT	Pembro. pneumonitis
L5	LLT	Immune-med. pneumonitis

Table 1: MedDRA hierarchy. Adverse-event mentions in clinical text map almost exclusively to L5 (≈96%).

NER conditions as a CCR perturbation..

Rather than comparing two NER systems, we use them as a controlled perturbation of CCR within the same pipeline. **Baseline NER** (ner_bio11m): BioMistral-7B zero-shot; CCR = 39.0%, 184

grounded spans across 472 extracted (126-segment evaluation set). This condition exhibits a systematic artefact—the model generates *hypothyroïdie* for 92/127 segments regardless of content—partially contaminating consistency measurements. **Fine-tuned (higher-quality) NER** (`ner_biollm_ft`): BioMistral-7B fine-tuned on a hierarchical MedDRA SFT dataset; 211 grounded spans across 580 extracted (36.4% CCR) with cleaner boundaries and no hallucinated terms. Graph-augmented systems benefit substantially under this condition, confirming that *effective* span quality (clean boundaries, usable grounding)—not raw graph retrieval—is the binding constraint once contamination is controlled.

4 System Architecture

Six systems form an ablation ladder (Table 2). S1–S5 share the same four-stage pipeline: French NER → MedDRA grounding via Neo4j → global terminology lock → MT decoding. S6 is a glossary oracle ablation that replaces the MedDRA graph with a curated FR→EN glossary extracted from the reference translations.

ID	Model	Graph	Enforce.
S1	NLLB-200 600M	No	None
S2	Mistral-7B (doc)	No	None
S3	Mistral + GraphRAG	Yes	None
S4	S3 + reranking	Yes	Soft
S5	NLLB/Mistral + logit	Yes	Hard
S6	NLLB + glossary	Oracle	Hard

Table 2: Ablation ladder. S1 = NMT lower bound; S2 = LLM with full document context; S3 is the key novel condition; S6 isolates term-source quality from graph retrieval.

Terminology planning.. Before translation, `compute_global_locks()` scores each candidate English rendering c :

$$\text{Score}(c) = \alpha \text{Ctx}(c) + \beta \text{Nbr}(c) - \gamma \text{HP}(c) \quad (1)$$

where Ctx is cosine similarity to source segments containing the term, Nbr is similarity to adjacent MedDRA concept renderings, and $\text{HP} = |\hat{d} - d_{\text{src}}|$

penalises wrong specificity level ($\alpha=0.5$, $\beta=0.3$, $\gamma=0.2$). When a term matches *multiple* MedDRA concepts, no lock is committed (§8).

5 Metrics

Fluency.. Corpus BLEU [Papineni et al., 2002] and chrF++ [Popović, 2017] via SacreBLEU against one professional English reference.

HTM and HTM-A.. For each segment containing a grounded French span, the hypothesis is scored against the MedDRA graph: $s_i = 1.0$ (exact level match), 0.5 (same branch, wrong specificity), or 0.0 (different branch / absent). $\text{HTM} = \text{mean}(s_i)$. HTM-A relaxes matching across three alias tiers: T1 exact LLT; T2 adds PT parent; T3 adds all same-branch ancestors. We compute **rHTM** by applying HTM to the professional reference as hypothesis, establishing a performance ceiling.

Hyp–Ref Agreement (HRA).. Fraction of HTM-measured segments where the system’s terminology choice matches the professional reference—a measure of human-like translation independent of MedDRA correctness.

CCR and drift.. $\text{CCR} = \text{grounded/extracted spans}$; identical across S1–S6 for a given NER condition. Term drift = $(D - 1)/(N - 1)$ for D distinct renderings across N segments; drift = 0 is perfect consistency.

6 Results

6.1 Main Results

System	BLEU	chrF++	COMET	HTM	HRA
S1 NLLB	22.89	37.14	0.639	0.321	0.672
S2 Mistral	22.51	39.95	0.637	0.413	0.743
S3 GraphRAG	10.39	34.36	0.586	0.397	0.601
S4 rerank	9.93	34.24	0.583	0.429	0.579
S5 NLLB+boost	21.59	35.91	0.638	0.353	0.661
S5 Mist+boost	10.32	34.32	0.585	0.380	0.607
S6 NLLB+gls*	20.00	35.11	0.634	0.321	0.694
Reference	—	—	—	0.151	1.000

Table 3: Baseline NER condition (`ner_biollm`, CCR = 39.0%; 126 segments, 48_028 excluded). BLEU/chrF++ via SacreBLEU; COMET via wmt22-comet-da; HTM/HRA from Neo4j. *S6 is a glossary oracle (terms from reference).

System	BLEU	chrF++	COMET	HTM	HTM-A
S1 NLLB	22.89	37.14	0.639	0.550	0.429
S2 Mistral	22.51	39.95	0.637	0.735	0.476
S3 GraphRAG [†]	19.58	35.33	0.626	0.664	0.440
S4 rerank [†]	19.66	35.36	0.626	0.664	0.450
S5 NLLB+boost	21.37	36.06	0.637	0.578	0.403
S5 Mist+boost [†]	20.61	35.78	0.626	0.673	0.450
S6 NLLB+gls [*]	20.09	34.81	0.632	0.536	0.408
Reference	—	—	—	0.178	1.000

Table 4: Fine-tuned NER condition (ner_biollm_ft; CCR = 36.4%; 126 segments, 48_028 excluded).

[†]Contamination-filtered (18/127 rows scored as empty hyps). ^{*}S6 uses a glossary oracle from the reference.

6.2 CCR is the Binding Constraint

Table 5 isolates the effect of NER quality by holding all other pipeline components fixed and comparing the same system across NER conditions.

System	Δ BLEU	Δ chrF++	95% CI Δ BLEU
S1 NLLB	± 0.00	± 0.00	[0.00, 0.00]
S2 Mistral [‡]	± 0.00	± 0.00	[0.00, 0.00]
S3 GraphRAG	+7.07	+0.97	[+1.07, +11.10]
S4 rerank	+7.42	+1.12	[+1.18, +11.53]
S5 NLLB+boost	−0.16	+0.15	[−0.98, +0.66]
S5 Mist+boost	+7.16	+1.46	[+1.10, +11.66]
S6 NLLB+gls [*]	+0.09	−0.30	—

Table 5: Paired gain (fine-tuned minus baseline NER) on 126 segments. Δ BLEU is the paired bootstrap estimate (2,000 resamples, seed 42); 95% CIs for S3–S5-Mistral exclude 0. Δ chrF++ is the corpus-level paired difference. [‡]S2 shares the same inference JSONL across conditions. ^{*}S6 CI not computed (oracle glossary).

S1 is unchanged (no graph). S2 is exactly zero: the same inference JSONL is shared across both NER conditions, so Δ BLEU is zero by construction. S3–S5-Mistral show ≈ 7 -point Δ BLEU with intervals strictly above zero, isolating the NER bottleneck on the graph path. S5-NLLB is effectively flat. The S6 oracle glossary yields negligible gain, showing that logit boosting with reference-extracted terms does not improve over the graph path once NER quality is matched.

Graph-free systems (S1, S5-NLLB) dominate on corpus BLEU and chrF in the baseline NER condition (Table 3) because the graph influences fewer than 40% of adverse-event spans while in-

ferred MedDRA strings still hurt fluency where the graph is active. The contrast between zero Δ BLEU for S1, S2, and S5-NLLB versus large Δ BLEU for S3–S5-Mistral in Table 5 isolates the bottleneck upstream of graph retrieval on the Mistral-augmented branch. Higher span coverage with cleaner NER would likely reverse the S2/S3 ranking.

6.3 Missing Terms, Flattening, and Consistency

Under baseline NER, S5-Mistral achieves the lowest missing-term rate; S1-NLLB misses the majority of grounded terms. Graph systems benefit from cleaner NER: S3’s missing-term rate falls substantially between conditions. Zero concept flattening events are observed (all grounded L5 source terms produce L5 hypotheses), partly an evaluation artefact—HTM cannot detect flattening unless upper-level aliases are provided, a limitation HTM-A addresses.

Mean cross-document modal agreement, averaged over French terms with ≥ 2 segments in error_analysis/, is highest for graph-free systems under baseline NER (S1: 0.993; S2: 0.984) while S3 scores 0.946; under fine-tuned NER, S3 rises to 0.976, surpassing S1 (0.970) but remaining below S2 (0.985)—so the “graph hurts consistency” pattern is clearest under noisy baseline NER and partially attenuates when NER improves, matching the missing-term and BLEU deltas.

7 The HTM–Practice Gap

7.1 Why the Reference Scores Near Zero

On the baseline NER manifest, rHTM = 0.151 on the professional reference (Table 3); under fine-tuned NER the reference HTM is 0.178 (Table 4) because the grounded span set differs. In both cases the score stays below several MT systems on HTM. Professional translators do not transcribe MedDRA LLT strings verbatim: the LLT *Immune-mediated pneumonitis* appears as *pneumonitis* in tabular contexts and as *immune-related pneumonitis* in descriptive contexts.

7.2 Alias Expansion Does Not Close the Gap

The ancestor chain for *pneumopathie inflammatoire* contains *pneumonitis*, so the null T1→T3 delta is not a graph coverage gap. Professional translators use phrasings such as *immune-related pneumonitis* and *inflammatory lung disease* that lie entirely outside the MedDRA preferred-label tree at any level. The mismatch is structural: MedDRA labels are designed for pharmacovigilance coding, not regulatory prose.

Tier	Accepts	Low	Hi
T1	Exact LLT	0.151	0.178
T2	T1 + PT parent	0.151	0.178
T3	T2 + branch ancestors	0.151	0.178

Table 6: rHTM-A on the professional reference under three alias tiers (Low = baseline NER manifest; Hi = fine-tuned NER manifest). All tiers yield identical scores within each manifest.

HTM should therefore be interpreted as a measure of MedDRA-proximity, not translation quality. The HRA column is the complementary measure. Notably, graph-augmented systems score consistently *lower* on HRA than S2 (Table 3: S3 = 0.601 vs. S2 = 0.699; Table 4: S3 = 0.440 vs. S2 = 0.476): the graph pushes outputs toward MedDRA labels that professional translators avoid.

8 Consistency Failure Analysis

Term drift (qualitative).. *Pneumopathie inflammatoire* occurs in four baseline segments where it is grounded for S3/S4/S5-Mistral, and in eight fine-tuned segments (more NER hits for the same French string). Those systems emit two renderings—often *pneumonitis* in incidence-style sentences and *immune-mediated pneumonitis* elsewhere—yielding drift scores of 1/3 (baseline) and 1/7 (fine-tuned). S1 collapses to one rendering for this span in both conditions (drift 0). S2 is stable in baseline (drift 0) but shows non-zero drift under fine-tuned NER once the span appears in additional contexts. S3/S4/S5-Mistral thus exhibit the clearest cross-segment inconsistency for this exemplar.

Mean modal agreement (§6) shows the same pattern at corpus level: graph-free systems rank highest under baseline NER; graph-augmented systems recover partially under fine-tuned NER but do not overtake S2.

Mechanism.. *Pneumopathie inflammatoire* maps to two distinct MedDRA concepts in the string cache (one of 7,679 ambiguous normalised keys). The planning stage sets `has_planning_lock = 0` for any ambiguous term and commits no lock. S3/S4/S5-Mistral then resolve the ambiguity context-locally per segment from RAG context: incidence-focused segments attract *pneumonitis*; mechanism-focused segments attract *immune-mediated pneumonitis*. S1-NLLB produces zero drift not because it reasons about

terminology, but because its fixed vocabulary has no mechanism for context-dependent selection.

This failure is independent of CCR. Even with perfect NER recall, any term matching multiple MedDRA concepts will be left unlocked, producing potential drift in all Mistral-based graph systems. The fix is *soft ambiguity resolution*: commit the top-scoring candidate as a provisional lock flagged for human review, consistent with CAT-tool workflows.

Output contamination.. In the fine-tuned NER condition, 18/127 segments per system (14.2%) contain prompt contamination: Mistral reproduced MedDRA metadata strings and instruction markers from its own system prompt. These rows were replaced with a null hypothesis for metric computation. Contamination biases raw scores in opposite directions—deflating BLEU (precision destroyed by injected strings) while inflating chrF (character overlap increased)—confirmed by S3 raw vs. filtered: BLEU 15.36→19.58, chrF 37.20→35.33. S1 and S2 are unaffected as they receive no graph context in their prompts.

9 Discussion and Limitations

The results collectively show that the TermPlanMT pipeline is architecturally sound but CCR-limited. Graph-augmented systems improve substantially when supplied with cleaner, higher-coverage NER input. The absence of BLEU movement for S1 (and near-zero change for S5-NLLB), together with large Mistral-graph Δ BLEU in Table 5, is direct evidence that the bottleneck lies upstream of graph retrieval on the graph-augmented path.

The S6 oracle glossary (terms extracted from the reference via an LLM) yields only marginal BLEU improvement over S5-NLLB (+0.09 points), suggesting that the quality of the terminology source matters less for NLLB decoding than the NER coverage upstream.

The HRA-HTM inversion—graph systems score higher on HTM yet lower on HRA—reveals a real tension: optimising toward MedDRA formal labels diverges from how certified translators actually write. For regulatory MT in practice, HRA is the more meaningful target.

Limitations.. *Single document and reference.* All results derive from one SmPC section against one professional translation, limiting generalisability. *HTM ceiling.* rHTM is low in absolute

terms (0.151–0.178 across NER manifests) and invariant to alias expansion within a manifest, so HTM is only interpretable relatively. *NER artefact*. The baseline NER condition contains hallucinated spans (*hypothyroïdie* in 92/127 segments). *Contamination*. Fine-tuned-condition Mistral-based graph systems exhibit 14.2% prompt contamination; filtered scores are used throughout. *S6 oracle*. The glossary is extracted from reference translations via Ollama; genuine oracle quality depends on extraction recall.

10 Conclusion

TermPlanMT demonstrates that CCR is the binding constraint on graph-augmented regulatory MT: at low coverage graph-free baselines (S1, S5-NLLB) dominate on BLEU and chrF, while improving NER quality yields ≈ 7 -point corpus BLEU gains for Mistral graph systems with bootstrap intervals excluding zero. Three broader findings emerge: MedDRA nomenclature and professional translation practice are structurally misaligned in ways that alias expansion cannot fix; ambiguous grounding produces predictable term drift through a mechanism fully explained by the planning stage’s abstain-on-ambiguity policy; and consistency metrics must control for system type, since NLLB’s apparent consistency is an artefact of its fixed vocabulary. The glossary oracle (S6) shows that replacing the MedDRA graph with reference-extracted terms provides negligible benefit, pointing to NER coverage rather than term-source quality as the primary bottleneck. Substantially higher CCR with cleaner span boundaries would likely be required to reverse the S2/S3 fluency ordering; quantifying that threshold is left to future work with broader French biomedical NER.

Acknowledgements

MedDRA licence held under non-commercial academic agreement via MSSO/UPV/EHU, supervised by Gorka Labaka.

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